# Xingjian Zhang

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## Education

**École Polytechnique, Institut Polytechnique de Paris** — France
PhD. candidate in Bioengineering — 2023 - Now
M.S. in Biomechanics and Biomedical Engineering — 2021 - 2023
B.S. in Computer Science and Mathematics — 2018 - 2021

## Publications

**Peer-reviewed articles**

1. *Modeling and predicting second harmonic generation from protein molecular structure*, Physical Review X, 2024
   B. Asadipour, E. Beaurepaire, X. Zhang, A. Chessel, P. Mahou, W. Supatto, MC. Schanne-Klein, C. Stringari,
2. *Interpretable exact linear reductions via positivity*, Computational Methods in Systems Biology, 2021
   G. Pogudin, X. Zhang

**Pre-prints**

1. *BioImageLoader: Easy handling of bioimage datasets for machine learning*, 2023
   S. Lim, X. Zhang, E. Beaurepaire, A. Chessel

## Experience

**Doctoral Researcher @ LadHyX, École Polytechnique | LTCI, Telecom Paris** — Nov 2023 - Now
- Research on dynamic cellular nuclear deformations on microgroove substrates to develop a functional in-vitro diagnostic tool for laminopathies and breast cancer.

**Deep Learning Intern @ Dassault Systemes** — Mar 2023 - Sep 2023
- Deep learning research in 3D tumor segmentation. Designed and developed AI models for the TWINONCO project.

**Research Intern @ LOB, École Polytechnique** — Apr 2023 - Mar 2023
- Image processing to segment axons and dendrites of mouse cerebellum images from THG microscopy. Developed segmentation tools for noisy 2D THG images and validated on 3D Imaris tracings.
- Modeled muscle fiber orientations of myosin/collagen/tubulin in zebrafish images from pSHG microscopy. Developed an algorithm to compute in-plane and out-of-plane fiber angles using structure tensors in Python. Automated Matlab code to compute anisotropy factor of different harmonophores.
- Development of Python library bioimageloader for wrapping bioimage datasets in a unified interface for machine and deep learning applications.

**Research Intern @ LTCI, Telecom Paris** — May 2023 - Jun 2023
- Deep learning on various datasets of different imaging modalities such MRI, CT, CXR etc. with different windowing settings to study the optimalities of VOI prior in full acquisition dynamics.

**Research Intern @ LIX, École Polytechnique** — Sep 2020 - Apr 2021
- Created an algorithm in julia using Fourier-Motzkin elimination to improve the physical interpretability of exact linear reduction for rule-based biochemical systems

## Awards

PhD Full Fellowship from École Polytechnique, IP Paris
Master's Scholarship from École Polytechnique, IP Paris

## Skills

**Programming:** Python, Julia / R, Matlab / C/C++, Pytorch and Multi-GPU learning
**Languages:** Chinese (native), English (proficient), French (proficient)